

MAAM: Anchor-Preserving Compression and Contextual Calibration for Chinese Discriminatory Language Detection

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Key idea: *preserve discrimination-relevant semantic anchors under structured compression, then calibrate predictions with C–I–S contextual priors.*

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Seeking Fall
2027 PhD!!!



1 · Why this problem?

Chinese discriminatory language is often **implicit and context-dependent**. Binary or coarse labels miss distinctions between overt hostility, subtle stereotyping, and emotional intensity. MAAM challenges the full-input assumption: not every token is equally informative.

2 · What is new?

01 MAAM

A plug-and-play framework combining anchor-preserving compression (Myopia) with C–I–S prior calibration (Astigmatism).

02 ChLGBT

A fine-grained Chinese LGBT-focused discriminatory-language dataset with three-dimensional annotations.

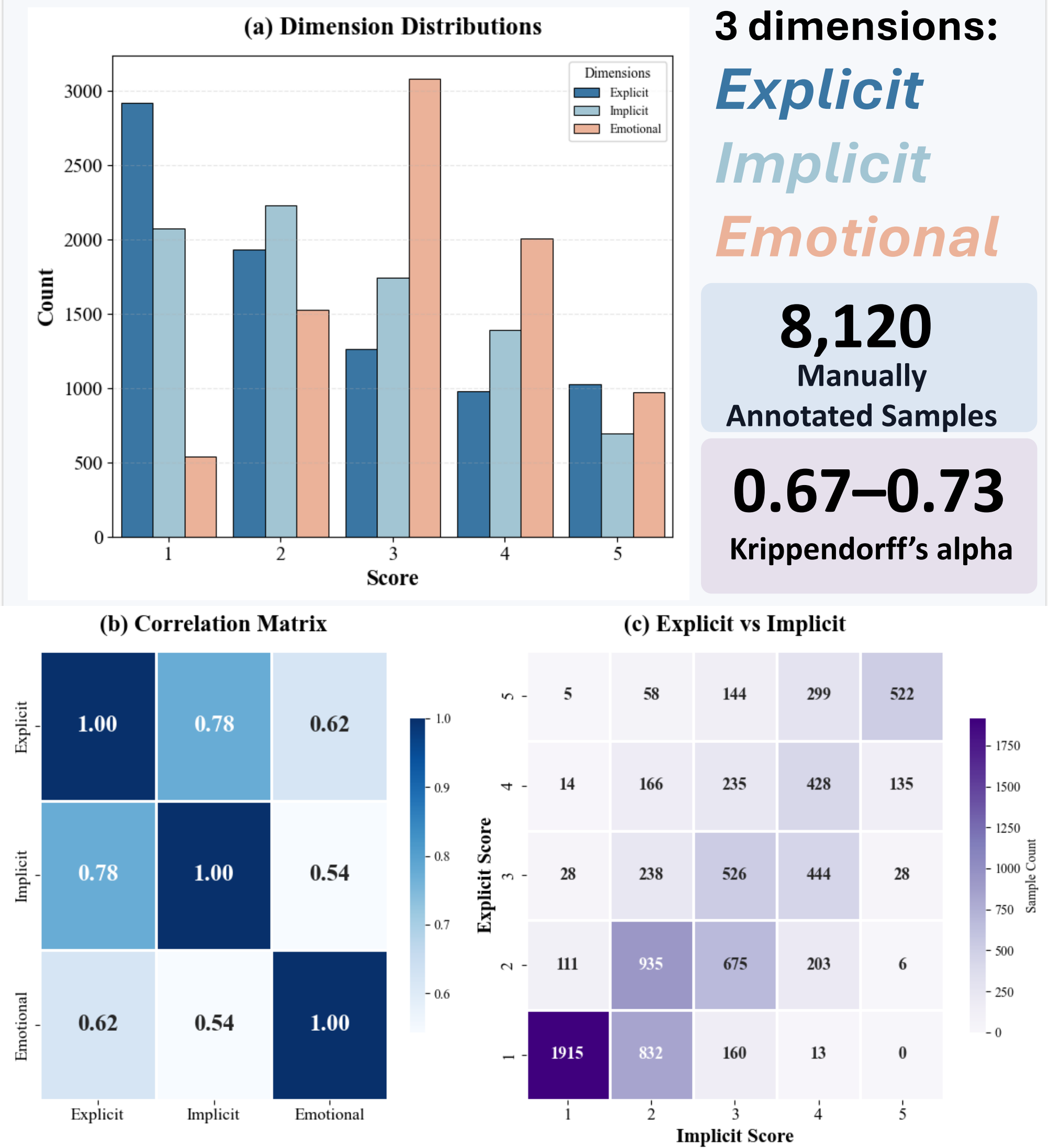
03 Generalization

Improvements extend beyond ChLGBT to coarse-grained COLD Region/Race tasks.

3 · ChLGBT

A fine-grained Chinese LGBT-focused discriminatory-language resource with three parallel 1–5 ordinal labels.

Collection: Weibo posts from Jan–Dec 2024. Original posts only; identifiers, profile metadata, source URLs and raw social-media texts are not released.



Why separate Explicit and Implicit:

“Monogamy protects ordinary people’s marriage and reproductive rights; without marriage-law protection, a gay man could keep multiple partners.” (**Explicit = 2, Implicit = 4.**)
No slur, no attack — yet it implies bias through a stereotype. A binary label would call this “neutral.” So overt and implied bias must be modeled separately.

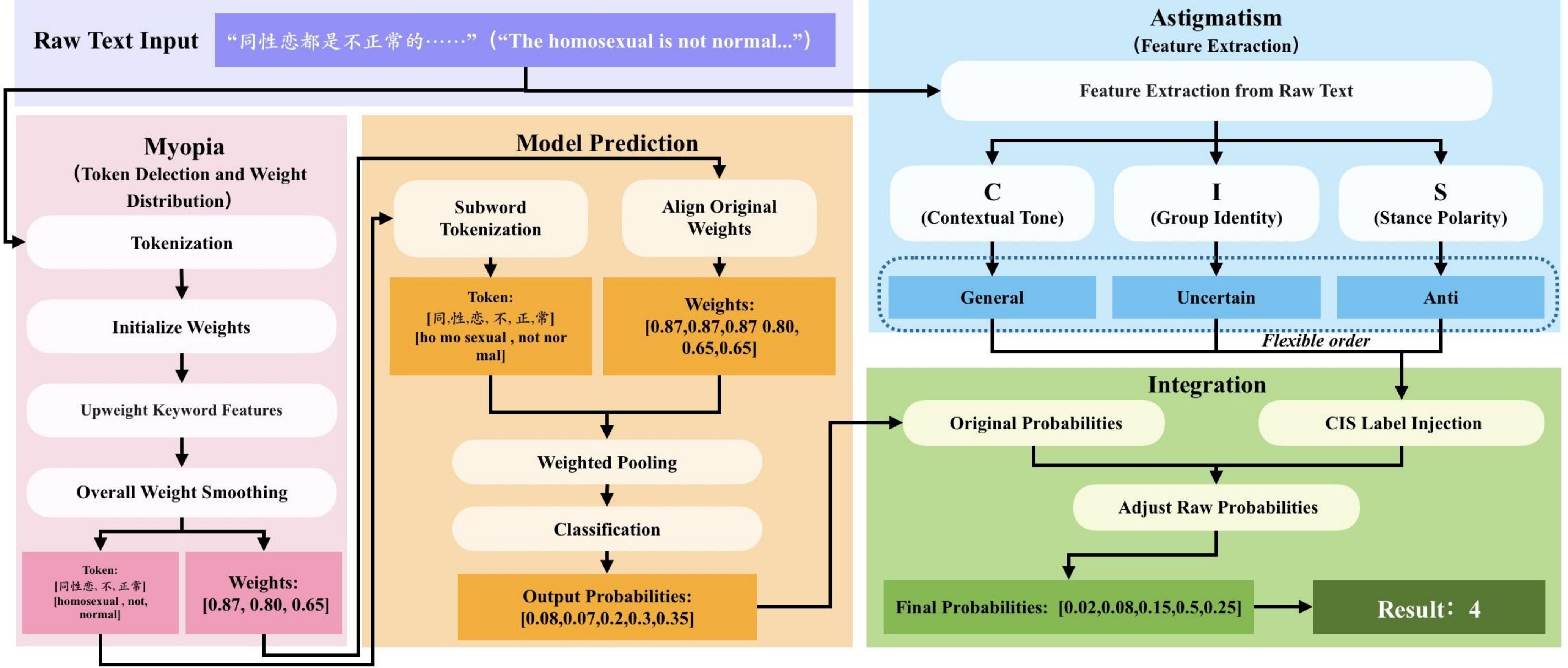


- Why MAAM exists
- Personal story of MAAM
- 5 universal principles
- Extend to any language / task / model
- MAAM + other models
- ChLGBT dataset access
- Author updates

4 · MAAM framework

Myopia → preserve anchors before encoding

Astigmatism → calibrate predictions after encoding



MYOPIA

Anchor-preserving compression

Layer	Function
1 Core Arguments	Who
2 Attributive Modifiers	What attribute
3 Predicative Events	What action
4 Modality & State	What modality
5 Contextual Frames	What context
6 Logical Connectors	What logic
7 Quantificational Details	How much
8 Punctuation Markers	What punctuation
9 Modal Particles	What tone

ASTIGMATISM

Contextual calibration with C–I–S priors

Extract three contextual signals:
C = Contextual Tone (*Funny, General, and Serious.*)
I = Group Identity (*Ingroup, Outgroup, and Uncertain.*)
S = Stance Polarity (*Pro, Anti, and Neutral.*)
Fuse their priors with model probabilities at decision time.

1. Tokenize and initialize linguistic weights.
2. Upweight keyword / identity features.
3. Smooth weights and suppress low-value tokens.
4. Filter level-5 tokens, then use weighted pooling.

5 · Results

MacBERT ablation + backbone transfer

Variant	Exp F1	Imp F1	Emo F1	BS	ECE
Baseline	0.34	0.32	0.32	0.737	0.082
w/o Myopia	0.45	0.44	0.42	0.686	0.077
w/o Astigmatism	0.40	0.37	0.44	0.675	0.076
MacBERT+M AAM	0.46	0.45	0.47	0.622	0.066
RoBERTa + MAAM	0.47	0.46	0.48	0.611	0.059

Full MAAM improves all three F1 dimensions while reducing Brier score and ECE.

Cross-domain generalization

Variant	F1	BS	ECE	F1	BS	ECE
COLD				Region		Race
Baseline	0.876	0.124	0.119	0.906	0.074	0.076
MAAM	0.881	0.097	0.071	0.928	0.052	0.054

TAKEAWAY

Robust discriminatory-language detection does not necessarily require a larger backbone.

MAAM preserves task-relevant anchors, adds contextual priors, and improves both prediction and calibration.

Zero-shot LLM comparison

Model	Exp F1	Imp F1	Emo F1
ChatGPT-4o	0.42	0.42	0.44
LLaMA-3.3	0.35	0.36	0.40
ERNIE-5.0	0.35	0.41	0.47
DeepSeek-V3.2	0.42	0.42	0.46
Qwen3-Max	0.39	0.37	0.40
MAAM (Ours)	0.46	0.45	0.47

MAAM remains competitive with frontier LLM baselines while using a compact encoder-based pipeline.

Few-shot LLM comparison

As the number of shots increases from 1 to 10, LLM performance generally improves modestly; the best 10-shot F1 scores are 0.47 (Explicit, ChatGPT), 0.45 (Implicit, ChatGPT), and 0.51 (Emotional, ERNIE).

Case Study

Sentence: "Yuepan (fatty), it's quite interesting to see a marriage-fraud gay talking about integrity!" ("月半，骗婚的同性恋谈诚信挺有趣的！")

What MAAM changed:

Δ Weight Suppressed: 月 (−0.170), 半 (−0.015), comma (−0.036), ! (−0.305)
Δ Weight Amplified: 骗婚 (+0.866), 同/性/恋 (+0.859~+0.865), 谈 (+0.807)
Effect: Weight shifted from *rhetorical/functional tokens* (nickname, punctuation) to *discrimination-relevant semantic anchors* (marriage fraud, homosexuality), focusing the model on the core biased content.